The Shape Channel

A Relational Decompiler from Binary Rings to Executable Geometry and Material Response

Operational invariants, reader-relative closure, and the first quantitative polyethylene calibration

Driven by Dean A. Kulik

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Research synthesis / computational manuscript. This manuscript integrates the Nexus relation-vocabulary ledger (§§1–163), the October partner-ring and ideal-chain audits, and the subsequent §167 polyethylene shape-channel reduction. Computations cited as **RUNS** were performed in the supplied research record unless expressly designated as independent mathematical derivations here. This paper is a synthesis, not a new blind prediction or a claim that the physical universe has been fully derived from binary arithmetic.

Abstract

A recurring limitation in computational descriptions of physical systems is the placement of objects before the transformations by which they become distinguishable. The Nexus programme reverses this order. It studies finite distinctions, their reversible transformations, the relations among those transformations, and the readers that expose only selected consequences. The central object of this paper is the **shape channel**: not a second value, nor a stored history, but the operational relationships exposed by an operator acting on a boundary. We give a unified account of the partner ring's fixed-point-free involution, the exact difference/face decomposition of binary words, representation-theoretic reader spectra, relational deficiencies and closure certificates, topology recovered from constrained token motion, and a geometry decompiler connecting ring operations to ideal planar and tetrahedral chains. The relation-vocabulary witness has 66 configuration vertices, 300 moves, and a 235-dimensional rational relation module; its 12 reader classes are evaluations of a six-component multiplicity vector rather than distinct fields. In the geometric realization, a word's XOR differences recover ideal dihedrals, the native partner reverses the ordered chain shape, and suffix involutions implement rigid bond-axis half-turns. In three-dimensional tetrahedral geometry, the compiler distinguishes Boyer's kinematically locked three-bond exchange from Schatzki's exact crankshaft and corrects the connected seven-bond class count to 16 of 38. The shared integral boundary operator $(-1,1)$ connects binary XOR and real bond displacement without identifying

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their values. Finally, a polyethylene rotational-isomeric-state transfer reduces from a 72-state orientation reader to four scalar equations, producing a characteristic ratio $C_\infty = 6.229676328$ at 413 K for declared literature-based energies, while independently reported estimates near 7–8 expose a genuine remaining constitutive gap. The work establishes a usable methodology for diagnosing hidden degrees of freedom, predicting executable local motions, reducing physical models by symmetry, and designing independent tests. It does **not** establish a unique matter law from 1,0 alone. The operative thesis is that **the in-between—the composition, boundary, and readout relation—is the transferable mathematical content across domains**.

Keywords: Nexus Framework; shape channel; reversible compiler; relational invariants; finite group representations; reader observability; graph braid groups; polymer kinematics; rotational isomeric state; molecular conformation; constraint decompilation.

1. The question: why calculations meet when histories differ

The elementary equality $1 + 4 = 2 + 3 = 5$ distinguishes a result from the routes that produce it. Addition identifies a common consequence; it does not specify why the incoming terms arose. An observable may similarly identify several histories, geometries, or material mechanisms. The mathematical problem is not to search backward through all possible histories until one fits. It is to determine which distinctions the readout necessarily retains, which it necessarily forgets, and what complementary interface would recover the forgotten structure.

The present approach is grounded in a recurring idea in Kulik's papers: **methods are shapes**. Here, *shape* is defined operationally. It is not restricted to visible Euclidean outline. It is the collection of relations that constrain what an operation can do and what another operation can distinguish. A spring, an electrical oscillator, and a local molecular mode may have different substances and parameter values, yet share a response algebra. That algebra is the point of contact. A cross-domain match must preserve an explicit operation or relation, not simply a numerical coincidence.

This study consequently separates four layers. The **state inventory** specifies the available configurations; the **route algebra** specifies reversible transformations and their compositions; **admissibility** restricts which operations can be executed in a particular realization; and the **reader** specifies which relationships become observable. Confusing these levels was the source of several earlier mistaken conclusions in the Nexus audit. The corrected distinction is central to this paper.

The programme's wider foundational language is Co/C1—absence of a privileged external change-maker and the insistence on continued distinction. In the finite mathematics below, those ideas motivate the design but do not prove a unique cosmic dynamical law. The claims subject to proof are instead concrete: involution identities, rank formulas, algebraic correspondence maps, kinematic feasibility under stated constraints, and quantitative transfer calculations.

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2. Construction and validation protocol

2.1 WHAT, WHERE, route and reader

An admissible **WHAT** is a declared transformation with its reverse or recovery interface. A **WHERE** is the address at which a specified reader returns its consequence. A **ROUTE** uses known compositional structure to reach an address. A **SEARCH** explores unknown addresses or guesses substitutions. The difference is methodological: deriving a missing transformation from a quotient, stabilizer, kernel or boundary is not equivalent to trying possible answers until one lands.

A missing component is classified by its dependency structure. **BOUND** means one established provider supplies the relation; **AMBIGUOUS** means distinct providers remain compatible with the existing observation and another independent readout is required; **UNBOUND** means no existing relation supplies the needed operation. In the research ledger, **STANDS** denotes an independently established theorem or measured fact; **RUNS** denotes a reproducible executed derivation or predeclared calculation at its stated scope; **DEAD END** means a calculation returned a different owner's structure; and **OPEN** denotes a missing proven link. Data not present in the archive are **UNLINKED**, not supplied by convention.

For a genuinely prospective result, the prediction text, its SHA-256 seal, and its return/readout must precede linking to target data. Several comparisons in this paper, including polyethylene's characteristic ratio, are explicitly retrospective and are not presented as blind predictions. Re-running two implementations of the same physical assumptions is a strong code check, but is not independent empirical corroboration.

2.2 Evidence hierarchy

Three different achievements must be distinguishable in any claim of novelty: (i) **mathematical identity**, whose proof is intrinsic and may be classical; (ii) **computational witness**, whose exact counts or constructions may be new to this specific substrate; and (iii) **physical landing**, in which a derived prediction is compared with measurements not used to select it. A fourth category, **interpretation**, proposes how the first three might form a more general ontology. This paper explicitly labels interpretations as such.

3. The partner ring: an addressable algebra of distinctions

3.1 Fixed-point-free pairing

For $x \in B_n = \{0,1\}^n$, let Jx reverse bit order and Cx complement every bit. Define the native partner

$$P_n(x) = \begin{cases} Jx, & Jx \neq x, \\ Cx, & Jx = x. \end{cases}$$

The construction has $P_n^2 = I$ and no fixed points for $n \geq 1$. Thus B_n is partitioned into 2^{n-1} opposed pairs. The prescribed ring order places canonical pair representatives and their partners symmetrically, including the special center-seam exchange. If $R_n[i]$ denotes the word at address i, then

$$P_n(R_n[i]) = R_n[2^n - 1 - i], \qquad P_n T P_n = T^{-1},$$

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where T advances one position in the constructed ring. The second identity is the dihedral reversal relation between partnering and successor traversal. It turns a word-level reversal/complement rule into an address-level operation with a direct return. The §135 audit reconstructed the ring and independently matched its address decoder at tested widths, including selected 64-bit addresses [1].

The ring does more than count all words. It supplies *where* opposite relations sit, how traversal reverses, and which nested transformations preserve a local boundary. This remains a finite address algebra. Addressability of 2^{64} words is not enumeration, nor is it evidence that every physical program or state occurs in nature.

3.2 Value and difference are complementary coordinates

Write $x = (b_0, \dots, b_{n-1})$ and define

$$\delta(x) = (b_0 \oplus b_1, \dots, b_{n-2} \oplus b_{n-1}), \quad f = b_0.$$

Then the inverse is explicit:

$$b_i = f \oplus \oplus_{j=1}^{i} \delta_j.$$

Consequently, $x \leftrightarrow (f, \delta)$ is a bijection. The difference word alone collapses precisely the pair $\{x, Cx\}$; the face bit selects a member of that pair. In other words, compression is reader-dependent and reconstruction is not magic: the inverse requires the omitted coordinate.

The ring's own partner satisfies the all-width identity

$$\boxed{\delta(P_n x) = \mathrm{rev}\big(\delta(x)\big).}$$

For a nonpalindrome it follows by reversing adjacent comparisons; for a palindrome, complement preserves all adjacent XORs and the difference word is palindromic. This is an algebraic **shape-channel identity**, not a resemblance between pictures [2].

3.3 Scaling and nesting

At width n, $|B_n| = 2^n$; growing to $n + 1$ exposes further distinguishable addresses without requiring a new primitive symbol. The October audit also documented a center nesting relation $C_{n+2} = 1C_n1$ and finite checks of a growing enclosure band, with selected high-width boundary tests [3]. Those tests warrant calling the pattern an observed ring construction, not asserting a fully proved all-width band-length theorem. The distinction between a theorem, an exhaustive finite check and a sampled 64-bit read must be preserved.

The Möbius-twisted clock in the record has $\tau^n = C$ and $\tau^{2n} = I$, with orbit-length structure governed by odd divisors of n on the declared word action [1]. Under the complementary-word quotient, a half-return may already read as closed although the full state has not returned. This demonstrates precisely how **observable closure** can differ from **state closure**.

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4. Operational shape: the invariant lives between the inputs

4.1 Boundary behavior, not a second payload

A source paper defines the shape channel as behavior exposed when an operator reads a value's boundary, rather than a second value stored beside the first [4]. This definition excludes the interpretation that every output must carry a chronological logfile. Instead, an operator O and reader R reveal a particular relationship $R(Ox)$ or comparison between such reads. Different operators can reveal different structural properties of one and the same underlying configuration.

A rigorous cross-domain match therefore seeks maps F and G for which the operational diagram commutes:

$$F \circ T_{\text{ring}} = T_{\text{phys}} \circ F, \qquad R_{\text{phys}} \circ F = G \circ R_{\text{ring}},$$

on the declared subset and with the declared observation interface. The mapping need not identify individual binary values with continuous physical coordinates. It may identify an incidence operator, a representation type, a symmetry action, a boundary condition, a homology class, or a quotient.

Such a diagram is **meaningful only when the arrows are specified**. Equality of one endpoint is weaker than equivalence of operators. A shared adjacency graph is weaker than a matching continuous constrained-motion space. A matching linear response equation is weaker than deriving the material constants in it. These distinctions define a grading of evidence instead of an all-or-nothing theory claim.

4.2 The smallest bridge: one integer incidence operator

The oriented bond boundary is the integer row

$$B = (-1 \quad 1).$$

Over $\mathbb{F}_2$, $Bb = b_0 \oplus b_1$. Over $\mathbb{R}$, $Bx = x_1 - x_0$ measures extension or compression. This is not a field homomorphism from $\mathbb{F}_2$ into $\mathbb{R}$; it is a **shared integer relation evaluated in two different coefficient systems**. This precision matters: the values do not carry over, but the incidence relation does.

For an ideal two-site spring with stiffness κ,

$$U(x) = \frac{\kappa}{2} \parallel Bx \parallel^2, \quad F = -\kappa B^T B x, \quad B^T B = I - S,$$

where S exchanges the two sites. The projectors $(I \pm S)/2$ identify the center-of-mass and relative-motion channels. With equal masses m, their squared frequencies are 0 and $2\kappa/m$. For a three-site path, the normalized squared frequencies are $0,1,3$; for a triangle, $0,3,3$. These are standard graph-Laplacian and vibration facts, but their appearance through the **same boundary reader** provides an explicit domain map [5].

The spring/RLC correspondence is likewise an operational equivalence after energy normalization. Mechanics and a series RLC circuit both realize a damped two-dimensional oscillator generator, with $\omega^2 = \kappa/m$ or $\omega^2 = 1/(LC)$ and damping b/m or R/L. The ring did not fix m, κ, R, L or C. It supplied a relational grammar that those implementations share.

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5. The relation-vocabulary field: methods, reach and reader blindness

5.1 One finite witness, two slots

The foundational ledger builds a two-token configuration arena from a 12-vertex, 30-edge witness. Its arena has **66 vertices and 300 moves**. The enumerated elementary three- and four-cycle vocabulary contains **1,283 cells**, and the full rational relation space has dimension **235**. The full automorphism group is $G \cong C_2 \times A_4$, of order 24 [1,6]. These are witness properties, not constants of the universe.

A cell is an individual closed transformation relation. A **family** supplies a set of such relations and spans a submodule W_F in the ambient relation module A. The five families are commuting squares (*disj*), spectator squares (*spec4*), rotation squares (*rot4*), spectator triangles (*spec3*) and rotation triangles (*rot3*). At the full-symmetry invariant reader the ranks are respectively $13,8,7,10,2$, while the union saturates a 15-dimensional visible ceiling. At the unsymmetrized reader their ranks are $216,168,150,174,20$, and the union spans all 235 dimensions. A five-family inventory is therefore not five independent contributions.

Table 1. Method reach and relation closure on the 12-vertex witness

Family	Cells	Full-reader rank	Unsymmetrized rank
Commuting squares	291	13	216
Spectator squares	528	8	168
Rotation squares	264	7	150
Spectator triangles	180	10	174
Rotation triangles	20	2	20
Full vocabulary	1,283	15	235

Two minimal family vocabularies span the full ambient module: $\{\mathrm{disj}, \mathrm{spec3}\}$ and $\{\mathrm{disj}, \mathrm{rot3}\}$. Yet domination of a deficiency is only a necessary check. Both *spec4* and *rot4* satisfy the commuting-square deficiency inventory but fall one dimension short of closing it at a faithful reader. *spec3* and *rot3* close it. The gap is the position of the required copy inside a multiplicity space, not simply its abstract type.

5.2 The representation reader as a functional

For finite G over characteristic zero, the ambient module decomposes into rational irreducibles λ, and an invariant-channel read at subgroup H returns

$$\dim V^H = \sum_{\lambda} m_\lambda\,(V)\,\dim(\lambda^H) = \langle m(V), f_H \rangle.$$

Here $m(V)$ is the method's representation multiplicity inventory; $f_H(\lambda) = \dim(\lambda^H)$ is the chosen reader. This is a classical consequence of semisimplicity and character theory, recovered in the Nexus ledger through rank measurements rather than introduced as a starting axiom [1,7]. Its operational consequence is sharper than the slogan "unreadable is not absent": **a zero output must be assigned to a zero in the module, a zero in the method vocabulary, or a zero in the reader's visibility profile**.

The ambient rational multiplicity vector on this witness is

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$$m(A) = (15,4,14,4,24,36)$$

across $1^+, 1^-, 2^+, 2^-, 3^+, 3^-$. The initial 12-reader table contracts these six columns in different ways. At full G symmetry, only 1^+ is seen. The quotient $Q(\text{spec4} \mid \text{spec3}) \cong 3^+ \oplus 3^-$ has six dimensions yet is completely invisible to that reader. A subgroup as large as possible while retaining at least one fixed vector in *every* rational irreducible gives a cheaper, still DC-faithful reader: an order-two subgroup inside A_4, with **152 cochain coordinates and a 115-dimensional relation ceiling**. DC-faithful is not lossless: it still suppresses nontrivial frequency components within some sectors.

5.3 A deficiency is a module, not a missing scalar

For a declared method span $W \subseteq A$, define the missing relation class

$$D = A/W, \qquad m(D) = m(A) - m(W).$$

Multiplicity domination provides a type check for a proposed filler, but cannot alone certify $W + W_F = A$. The required certificate is a span equality, which can be checked at a reader that sees at least one invariant direction of every irreducible. The ledger reports that the commuting-square deficiency is $(2,0,1,0,2,3)$, while the rotation-triangle module equals that deficiency plus a single redundant 1^- copy. That copy accounts for a reader-dependent one-dimensional overlap, invisible when the central involution is averaged away [1].

The deeper reader correction comes from moving below the five-family granularity. A G-stable collection of 33 individual cycle orbits spans 232 of 235 dimensions but already fills all 15 full-G invariant directions. Its quotient is a single 3^- representation. Thus the invariant reader reports closure while three genuine relation dimensions are missing. This explicitly refutes the extrapolation that saturation of the most symmetric observable implies full structural closure. In the ledger's formalism, a **reader-saturated state is not necessarily a relation-closed state**.

5.4 The full spectrum and the interface width

For a cyclic read $H = \langle g \rangle$ of order r, invariant averaging selects only Fourier bin zero. The other bins are recoverable from the same group action by weighting the orbit with characters:

$$P_k = \frac{1}{r}\sum_{j=0}^{r-1} e^{-2\pi i k j/r}\, \rho(g)^j.$$

For the central involution, $P_\pm = \big(I \pm \rho(z)\big)/2$. On the witness the full relation field splits $235 = 115 + 120$, yet the ordinary invariant read sees only the first 115. In one measured minimal vocabulary, its single overlap appears wholly in the antisymmetric channel; a claim of independence based on the invariant count was therefore false [1]. The representation remains, but retaining only the invariant projection **really discards** the other output; "nothing was destroyed" is true only of the full decomposition, not of a stored projection.

The vocabulary–reader measurement table is bilinear. Its rank is **six**, the number of rational irreducibles that couple the two sides on this witness. This resembles reachability–observability factorization in realization theory. The analogy is useful because it distinguishes a method that never reaches a direction from a reader that cannot observe a direction, and because changing both slots can lower the realized interface rank below

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either slot's standalone rank. The numerical width six is not a universal Nexus constant; it belongs to the declared witness and channel basis.

6. Closed encounters produce topological structure

The reader-blind portion of the two-token witness did not turn out to be an empty or structureless set. Its 24 reader-internal moves lie on a 12-vertex, degree-four graph. The 14 short relations on those moves comprise eight triangles and six squares. An explicit graph isomorphism identifies the graph as the **cuboctahedron**, with Euler characteristic $12 - 24 + 14 = 2$ and a one-dimensional top face relation [1]. The result is not a sphere created from nothing by the reader: it is the **medial graph of the octahedral core** already embedded in the original witness. Its 12 vertices correspond to the octahedron's 12 edges, its eight triangular faces correspond to old faces, and its six squares to old vertex figures.

This is a useful instance of a relational decompiler recovering a known geometric object *without supplying the word cuboctahedron as a generative instruction*. But it is an **inherited geometry** exposed by a token-motion operation, not a previously unknown topological theorem. Controls confirm limits: a degree-five icosahedral core loses the square faces under the three-/four-cell cutoff, while the tetrahedral case overcloses because additional short cycles are admitted.

An important correction prevents a false metaphysical conclusion. The 14 faces had zero weighted perimeter under one invariant price reader, but their individual boundaries could be read at degree one with a finer subgroup reader. Their zero arose from **reader invariance**, not a fundamental mismatch of cochain degree. The aggregate oriented sphere class has zero boundary by $\partial_2[\Sigma] = 0$, a separate Stokes/homology fact. These are two different mechanisms that earlier interpretations had conflated. A good decompiler must distinguish them.

The broader graph-braid studies in the ledger identify loops by their motion history, exchange class and route through the configuration arena, rather than by an endpoint alone. When topology distinguishes an exchange, merely writing the same terminal word does not certify that a collision-free path exists. This distinction becomes decisive in physical kinematics.

7. From ring differences to ideal molecular geometry

7.1 The planar honeycomb dictionary

Let a labeled planar honeycomb chain have unit bonds and $120°$ internal bond angle, encoded by a turn word x. The sign of consecutive local face normals changes exactly at positions where adjacent turn bits disagree. Hence the difference word $\delta(x)$ is the local cis/trans pattern (with the chosen normal convention). The face bit identifies the global alternative embedding. In this ideal model, the **ordered squared-distance matrix** of the chain is determined by $\delta(x)$, and equality of distance matrices implies equality of that difference word. A complete finite audit gave 8,188 exact word/shape checks across widths two through twelve [2].

The native ring partner identity now acquires a geometric interpretation:

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$$\delta(Px) = \mathrm{rev}\big(\delta(x)\big).$$

It acts on ordered distance matrices by reversing bead indices. This is a remarkable exact reader of **head-to-tail reversal of ideal chain shape**, not a rule that chemically different monomers may always be interchanged in actual polymer matter.

7.2 The geometric meaning of a suffix involution

Let D_k complement all turn bits at and after position k. Algebraically, for $k \geq 1$ it flips only the corresponding difference coordinate δ_k and leaves the others fixed. Geometrically, it is precisely a rigid $180°$ rotation of the downstream chain about the preceding covalent bond. The bond lengths and internal angles are preserved throughout that rigid half-turn. The §162 audit tested **18,432 exact coordinate comparisons** across widths two through ten [2].

One can interleave such operations to obtain a Hamiltonian Gray traversal of **every encoded turn word**. However, the full turn-word space contains self-intersecting configurations. On valid zero-thickness planar chains the Gray ordering itself is not a Hamiltonian cycle. A different finite result is positive: the graph including every valid single-suffix-half-turn edge was **connected at every tested width** $m = 2, \ldots, 12$. At $m = 12$, it contained 2,796 valid words and 14,728 undirected allowed edges; the prescribed Gray cycle contained only 2,739 valid-valid edges [2]. This is a finite-domain existence result, not an all-width theorem.

The operational lesson is specific: the ring may address a transformation, the geometric realization may permit its endpoint, and physical admissibility may still prohibit the path between those endpoints. State equality, endpoint congruence, differential mobility, and global continuous reachability must be certified separately.

7.3 Exact local rigidity from second-order compatibility

A five-bond global span bound originally proposed for a 12-cycle crankshaft was false and was withdrawn. Its replacement is an exact second-order argument. After factoring out the rigid-axis motion and denoting the four transverse perturbation coordinates by $u_2, \ldots, u_5$, the computed compatibility forms Q_1, Q_2, Q_3 satisfy

$$\begin{aligned} 2(2Q_1 + 9Q_2 + 13Q_3) = \quad & (u_2 + u_3 - u_4)^2 \\ & +6(u_3 - u_4 + u_5/3)^2 \\ & +9(u_4 - 5u_5/3)^2 + 13/3\, u_5^2. \end{aligned}$$

The positive leading principal minors are $1, 6, 54, 234$. Consequently, no nonzero sufficiently small transverse branch can satisfy the exact local constraints in the declared model: the local constrained motion consists only of the rigid crankshaft orbit. The result is **local rigidity**, not a proof that every globally disconnected configuration has been classified [2].

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8. The three-dimensional calibration: trans, gauche and a real crankshaft

8.1 Geometry supplied, mechanism recovered

The next compiler test uses the tetrahedral bond geometry of polyethylene-like carbon chains and the three local torsional states **trans**, **gauche+**, and **gauche−**. Given fixed neighboring bonds and angles, the relevant question is whether two compatible endpoint conformations lie on the same continuous exact-constraint component. This avoids naming a particular mechanism as a search candidate and instead classifies the methods permitted by the given local shape.

The October §163 audit reproduced the exchange counts over windows of two through six internal torsional bonds. The counts of elementary exchanges were **0, 12, 21, 213, and 981**, organized into **0, 4, 5, 38, and 175** symmetry classes, respectively [1]. The class count, not the number of endpoints, is the relevant catalogue unit. A window of k internal bonds with both exterior ends held is a $(k+2)$-joint linkage; the earlier use of a label one joint too short was corrected.

Table 2. Reproduced tetrahedral exchange inventory

Internal window k	Elementary endpoint exchanges	Geometric classes
2	0	0
3	12	4
4	21	5
5	213	38
6	981	175

8.2 What the geometry discriminated

The three-bond **Boyer-type** exchange $TGT \rightarrow TG'T$ is present among the short endpoint exchanges, but its exact held-exterior configurations are kinematically isolated. The endpoints cannot leave their positions by a continuous motion that keeps every bond length, angle and exterior boundary fixed. Similarly, exchanges of four or fewer bonds fail under the exact constraints, with two cases requiring a third-order test [1]. Actual thermally activated polymer rearrangement may still occur by bond-angle or length distortion and energy-barrier crossing; this calculation is a rigid-constraint verdict, not an absolute prohibition in real polyethylene.

The **Schatzki-type** sequence $TGTGT$ contains an executable rigid crankshaft. In the ideal eight-bead geometry the first and seventh bonds share an axis, and the intervening four beads rotate by **120°** into the next lattice position while exact bond lengths and angles remain fixed. The audit reports waypoint clearance no smaller than roughly one bond length in the declared measure; a finite waypoint certificate must not be promoted to a mathematically uniform collision bound without interval or continuation control [1,8]. The mechanism agrees with the classic polymer-motion literature and is a **calibration**, not the discovery of Schatzki's motion.

The seven-bond-class continuation count is **16 of 38 connected classes**, correcting the earlier 13. The three missed connections pass through an all-gauche helix where real branches cross. The distinction is operational: a continuation code that cannot restart at the crossing may declare disconnected states even

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though explicit branch-switched routes exist. The companion negative classifications were made using separately checked loop structure, and their scope remains the ideal articulated-chain model [1].

8.3 Singularities identify local flattening

All singular diamond-lattice endpoints found in the stated audit occurred at locally flat zigzags containing two consecutive trans twists: **48 of 48** in the seven-bond test and **36 of 36** in the eight-bond test. The pattern is **necessary in that finite census, not sufficient** for singularity. It connects the planar contact-degeneracy analysis to the three-dimensional polymer model: the first derivative can fail to see a transverse mode precisely where a second- or third-order compatibility calculation is needed. This is a testable method for diagnosing singular mechanism families without treating a vanishing Jacobian row as absence of a relation.

9. The nonlinear contact channel

In the ideal-chain contact example established earlier, the squared separation of two specified beads is

$$\| p_6(\theta) - p_1 \|^2 = 4 - 3\cos\theta.$$

The difference from its contact value is $c(\theta) = 3(1 - \cos\theta)$, with $c'(0) = 0$ and $c''(0) = 3$. If—and only if—the material model imposes the quadratic penalty $U = (\eta/2)c^2$, the leading response is

$$U(\theta) = \frac{9\eta}{8}\theta^4 + O(\theta^6).$$

Thus a first-order stiffness reader reports zero at this configuration while the next compatibility order supplies a quartic restoring term [5]. The shape relation predicts the **order of vanishing under that constitutive choice**, not the value of η, the physical energy functional, or a measured molecular potential. This is the correct scope for turning a geometric singularity into a material hypothesis.

10. A quantitative matter readout: polyethylene's characteristic ratio

10.1 The observable and its physical ingredients

The unperturbed long-chain characteristic ratio is

$$C_\infty = \lim_{N\to\infty} \frac{\langle \left|\sum_{j=1}^{N} \mathbf{b}_j\right|^2 \rangle}{N\ell^2} = 1 + 2\sum_{j\geq 1} \frac{\langle \mathbf{b}_0 \cdot \mathbf{b}_j \rangle}{\ell^2}.$$

This is a direct measurement target rather than a word-to-particle label. The shape grammar is the ideal tetrahedral chain and three torsion states. The material-specific input is their relative statistical weight, parametrized here by the trans/gauche energy difference E_g and the opposite-handed consecutive-gauche penalty E_p.

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Let $\sigma = e^{-E_g/(RT)}$ and $\omega = e^{-E_p/(RT)}$. The declared rotational-isomeric-state (RIS) relative transfer is

$$U = \begin{pmatrix} 1 & \sigma & \sigma \\ 1 & \sigma & \sigma\omega \\ 1 & \sigma\omega & \sigma \end{pmatrix}.$$

The Perron eigenvalue is

$$\lambda = \frac{1 + \sigma(1+\omega) + \sqrt{\big(1 - \sigma(1+\omega)\big)^2 + 8\sigma}}{2}.$$

The stationary transition is formed by the corresponding Doob transform, not by naively normalizing each row of U. Gauche reflection splits the full orientation transfer into even and odd components and reduces the original **72-state bond-pair/torsion reader to four scalar equations**. This is an exact reduction of the declared RIS model, not an exact reduction of all polyethylene physics [9].

10.2 Reduced shape-channel calculation

With the source's definitions

$$a = 1/\lambda, \quad q = (\lambda - 1)/(2\lambda), \quad r = \sigma/\lambda, \quad s = \sigma\omega/\lambda,$$

and $t = r + s$, $\delta = r - s$, $d = 1 - t$, define $\alpha = (2\delta - 1)/(2 - \delta)$ and $\beta = (1 - \delta)/(2 - \delta)$. The parity-odd coordinate is forced algebraically to be $w = \delta v/(2 - \delta)$. The remaining four coordinates obey

$$\begin{pmatrix} 1 & -a & 0 & -2q\alpha \\ -a & 1 & -2q & -2q\beta \\ 0 & -d & 1 & -t\alpha \\ -d & 0 & -t & 1 - t\beta \end{pmatrix} \begin{pmatrix} u_0 \\ v_0 \\ u \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

Writing $\pi_0 = d/(d + 2q)$ and $\pi_g = q/(d + 2q)$ yields

$$C_\infty = 2\left[\pi_0\left(\frac{u_0}{3} + v_0\right) + 2\pi_g\left(\frac{u}{3} + v\right)\right] - 1.$$

The reduction has a particularly useful exact control. When $\omega = 1$,

$$\boxed{C_\infty = \frac{2(2+\sigma)}{3\sigma}}, \qquad C_\infty|_{\sigma=1} = 2.$$

The equally weighted value two is the geometry-only baseline: $\langle \mathbf{b}_0 \cdot \mathbf{b}_j \rangle/\ell^2 = (1/3)^j$ and $1 + 2\sum_{j\geq 1} 3^{-j} = 2$. This baseline illustrates the direction of derivation—local relations first, material weights afterward.

10.3 Numerical results and the observed gap

The §167 computation used the declared illustrative values $E_g = 2.100$ kJ/mol and $E_p = 8.400$ kJ/mol. The independent 72-state calculation and four-coordinate reduced implementation agreed to approximately 10^{-15} [9].

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Table 3. Characteristic-ratio output of the stated ideal RIS model

Scenario	C_∞
All torsion methods equal	2.000000000
413 K, trans/gauche penalty only	3.124392391
413 K, both penalties	**6.229676328**
413 K, opposite-handed pair forbidden	6.715345190
300 K, both penalties	7.200650328

The previously published molecular-dynamics estimate for polyethylene at 413 K is approximately **7.9,** with historical viscosity-derived **6.8** and small-angle neutron scattering **7.8 ± 0.4** cited in the same source [10]. The §167 model's **6.2297** is therefore not agreement at the central neutron estimate. The source's energies and external benchmarks were known before the calculation: this is a retrospective physical calibration, not a presealed physical landing.

The source reports a 1,001-point monotone numerical sweep from $\omega = 0$ to 1 at fixed E_g; that finite sweep is not a proof of global monotonicity. Its extremal forbidden-pair value of 6.7153 is a useful model diagnostic but not, without such a proof, an exhaustive analytic upper bound on all interior parameter choices.

10.4 Reverse the gap: which measurement is missing?

One measured C_∞ cannot determine two unknown effective energies. At the declared 413 K point, the calculated local sensitivities (energies measured in kJ/mol) are

$$\frac{\partial C_\infty}{\partial E_g} = 0.790253691, \qquad \frac{\partial C_\infty}{\partial E_p} = 0.134369065,$$

so the local constant-ratio tradeoff is $dE_p/dE_g \approx -5.881217$. This is not a missing fitted constant: it is an **identifiability defect**—one scalar output for two material degrees of freedom.

The existing transfer already supplies a mathematically independent reader: the stationary trans fraction π_0. At 413 K it returns $\pi_0 = 0.596663987$, and the two-observable Jacobian is

$$J = \begin{pmatrix} 0.790253691 & 0.134369065 \\ 0.052463067 & 0.003102058 \end{pmatrix}, \qquad \det J \approx -0.0045980002.$$

The nonzero determinant removes the **local algebraic** ambiguity, although the inverse is ill-conditioned (condition number approximately 140 in the stated coordinates). An independent trans-fraction measurement under conditions comparable to the chain-size benchmark remains **UNLINKED**. A solid-state NMR study does measure gauche/trans populations, but the reported crystalline and lamellar effects make it an inappropriate plug-in calibration for the melt without an explicit state-of-matter bridge [11].

The next physical experiment is therefore not "try different energies until C_∞ matches." It is to obtain a second independent material readout, condition it consistently with the first, and test both against a prediction fixed before the data are consulted.

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11. What is new, and what is recovered?

Novelty must not be assigned to every recovered theorem. Character decomposition, Maschke's theorem, Fourier projectors, graph Laplacians, the Ho–Kalman realization viewpoint, conventional RIS statistics, and the historical Boyer/Schatzki mechanisms predate this programme. Their **recovery from one constraint-and-reader workflow** is a methodological convergence, not priority over the original mathematics or physical discoveries [7,8,10,12].

The contributions more specific to this work are: a finite partner-ring operational address system with palindrome-aware reflection and exact difference/face reconstruction; a measured relation-vocabulary **module ledger** that exposes hidden reader sectors and gives a closure procedure; a concrete cuboctahedral reader interior recognized as the medial graph of the witness core; explicit ring-to-ideal-chain identities including the native partner reversal; finite counts and corrections for valid half-turn graphs and tetrahedral exchanges; an exact crankshaft local rigidity certificate; and a symmetry reduction connecting the declared molecular grammar to a quantitative material observable. These results are best described as **new constructions, finite computational witnesses, and cross-domain synthesis in the Nexus research record** unless a comprehensive priority review establishes a stronger novelty claim.

Table 4. Claim scope and provenance

Result	Evidentiary class	Exact limit
$P^2 = I, PTP = T^{-1}$; invertible (f, δ)	Algebraic identities; finite regressions	Declared partner-ring construction
Six-channel multiplicity ledger, faithful-reader closure	Standard representation theory + witness-specific ranks	Finite rational G-modules
Pure 3^- deficiency under invariant saturation	Exact witness construction	Selected 33-orbit vocabulary
Blind cuboctahedral interior	Exact graph reconstruction	One $K = 2$ witness
$\delta(Px) = \mathrm{rev}\delta(x)$	All-width identity + geometric finite audit	Defined ideal chain geometry
Valid half-turn graph connected, widths 2–12	Finite exhaustive RUNS	Not proved for all widths
Boyer isolated; Schatzki executable; 16/38	Constrained tetrahedral computational model	Ideal bond/angle and boundary assumptions
72-to-4 polyethylene transfer	Symbolic reduction + numerical agreement	Declared RIS energy model
Complete physical law from $1,0$	**OPEN**	No independent derivation of constitutive constants

11.1 Other domain collisions: shells, register stacks, and exact phase routes

The present manuscript concentrates on the strongest recent **geometry-to-matter** branch, but the earlier ledger contains several additional relational crossings whose scope matters for understanding the emerging architecture. They do not automatically form a proof of one universal physics; they show how the same *method of locating a missing relation* has been applied in distinct mathematical languages.

In the atomic/nuclear shell programme (§§99–104), the recurrent question was why an orbital without a predecessor is admitted in a different order in atoms and nuclei. The ledger separated the two physical gates rather than assuming one shell-filling algorithm: the screened atomic centrifugal wall delays an orbital's entry into the core, while the nuclear spin–orbit splitting advances an aligned branch. From the nonrelativistic Thomas–Fermi model, its reported necessary atomic opening gate is $Z_{\mathrm{open}} \approx 0.1573(2\ell + 1)^3$, returning $4.2, 19.7, 54.0$ for p, d, f versus observed first occupations at $5, 21, 58$ [1]. This

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cubic scaling is a recovery of established Fermi/Thomas–Fermi physics, not a new universal constant. The Dirac mean-field audit subsequently recovered the sign reversal of the spin–orbit splitting from scalar/vector channel structure, but the origin of the nuclear scalar–vector potentials remained OPEN. Here "the piece with no way home" was a method of **typing the delayed orbital**, not proof that the ring alone generated the periodic table.

The rational-ring/BBP branch (§§119–133) treated digit-extraction lanes as phase-coupled orbits rather than independent series terms. In the reported eight-lane decomposition, the rational lane matrix had rank seven and a one-dimensional zero relation; the zero route came from cancellation across the specified phase and scale returns, not from absence of an operator [1]. A further correction was important: if two cyclic coordinates have orders h_1 and h_2, their actual lockstep orbit has $h_1 h_2 / \gcd(h_1, h_2)$ states, not always the full Cartesian product. A reader that discards the shared phase constraint changes the average. This is an exact **reader-and-context** issue in a number-theoretic setting, not evidence that every physical constant has been generated from a binary clock.

The SHA/register branch (§§105–114 and §§123–130) likewise distinguished full internal reversible transport from a compressed external readout. Its difference compiler implements the Boolean Möbius transform using successive XOR fusions; the transform is an involution under its defined coefficient convention. The same programme examined SHA round updates and BBP lane evaluations through relation/reader decompositions. These audits can verify a route, identify which context a reader discarded, and explain why a digest is a many-to-one projection. They do **not** establish general digest-only inversion or a polynomial-time preimage algorithm [1,3].

Finally, the graph-braid branch (§§73–75) treats token exchange as a class of **paths in a configuration space**, not a pair of positions. Its appearance alongside the ideal-chain move graph is not incidental: both insist that a proposed endpoint identification cannot substitute for a legal route. The common interface here is the topology of admissible continuation; the physical substrates and quantitative costs are different.

Taken together, these cases support an appropriately limited generalization: **the deficiency belongs to a particular operation and reader, not to the numerical output considered in isolation**. A shell gate, an orbital phase, a cryptographic carry and a polymer torsion are not the same physics. They can nevertheless be investigated using the same discipline of exact routes, retained contexts and separately stated readouts.

12. The emerging architecture: a relational decompiler

The combined result is neither a catalogue of preassigned particles nor a universal lookup table. It is more precise to call it a **relational decompiler**. Its input is an observed or prescribed local transformation domain. Its internal objects are relations: which states differ, which operations compose, which loops close, which boundaries are held, which quantities remain conserved, and which reader projects the result. Its outputs are constraints on the methods that can actually run and predictions about what specific readers should detect.

The distinction between a compiler and a decompiler here is directional. A forward construction produces all addresses permitted by a relational grammar; the reverse construction starts from an observed consequence and subtracts what the already established route accounts for. The residual is typed by kernel, quotient, stabilizer, representation content or compatibility degree. A good reverse route does not propose a random

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answer and test whether it resembles the output; it identifies which specific *kind of relation* must be missing, and whether the available structural owners bind it.

This clarifies Kulik's "potential outward, reality inward" language as a **working interpretation** of dual relations, not a verified physical wave equation. If K_n maps admissible routes between levels, forward possibilities propagate through K_n and boundary-compatible readouts pull back through its dual K_n^*. Their pairing agrees by adjunction. That dual accounting is mathematically sound; showing that these are actual physical propagating fields requires independently deriving the time evolution, locality, energy, and physical measurement law. The current results do not establish those elements.

The mature claim is narrower and more useful: **a shared operational relation can remain the same while its values, coordinates, trajectories, and physical realization change**. One does not have to map bits directly to carbon atoms to import a useful calculation. One must show that the maps preserve the relation and explain what extra physical information is required before the calculation predicts nature.

13. Practical uses of the shape-channel approach

13.1 Automated mechanism diagnosis

In a kinematic chain or articulated machine, construct the local constraint Jacobian, identify rank defects, distinguish trivial rigid motions from transverse modes, and compute higher-order compatibility where first order is blind. Use symmetry to reduce equivalent endpoint exchanges. The ideal polymer results give working examples of an endpoint exchange that cannot run and a crankshaft that can. A practical implementation would return the exact constraints, the proposed motion, its inverse, a local/continuous certificate, and the remaining collision risk.

13.2 Molecular conformational analysis

Begin with experimentally supported bond geometry and torsional states. Decompile which local exchanges are kinematically possible without bond or angle distortion; then attach independently measured energy weights to predict populations and chain observables. This prevents the common mistake of interpreting a geometrically valid endpoint exchange as a low-energy physical pathway. For polyethylene, the next implementation should jointly read C_∞ and a trans/gauche population, quantify sensitivity and model misspecification, and compare out-of-sample temperatures.

13.3 Instrument and sensor design

In a symmetry-rich system, a null invariant readout may conceal a nonzero antisymmetric or higher-representation component. Instead of increasing raw sensor count, calculate the representation content of the target deficiency and choose a reader with a nonzero response in that sector. Where the first linear reader is silent, a permitted nonlinear coupling may export part of the hidden channel, but its sign/phase ambiguity must be stated. This is a concrete observability strategy rather than a general appeal to "more resolution."

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13.4 Computational verification and compiler tooling

A route carries its own inverse or an explicit statement of lost information. Quotient maps disclose which differences were discarded and what extra context an inverse needs. Finite algebraic constraints can then be checked by exact arithmetic, while numerical geometry requires residual, Jacobian, and collision certificates. The same discipline applies to reversible internal rounds of cryptographic primitives, but **does not** turn a compressed digest into an invertible map. Reversing a SHA-256 round with full state is not solving an unrestricted SHA-256 preimage [1,3].

13.5 Structural search replacement

For a gap with a known type, compute the deficiency A/W before testing candidate fillers. In semisimple rational settings, its irreducible multiplicities are necessary method requirements; one faithful-reader span calculation determines whether a proposed family actually closes it. In physical parameter inference, the analogous step is to calculate the rank and kernel of the observable Jacobian before fitting. Both replace blind enumeration with a typed statement of what the existing measurements can and cannot determine.

14. A prospective empirical protocol

A complete physical test of this programme must be designed so failure is possible. For the first material domain, a credible test would freeze the exact 72-state and reduced four-coordinate readers, the torsion grammar, the temperature definition, the energy source, and the expected $C_\infty(T)$ and $\pi_0(T)$ readouts. The prediction and its way home would be sealed before consulting the selected experimental datasets. A second, independently obtained trans-fraction readout should be made at matched thermodynamic conditions. One dataset could be used to select or validate the constitutive weights, while another temperature or chain-length regime is withheld for prediction.

The missing physical law is not resolved merely by adding arbitrary terms until the characteristic ratio matches. Any extension—bond-angle flexibility, longer torsional memory, nonbonded interactions, or condensed-phase constraints—must declare which measured discrepancy requires it and what new observable distinguishes that extension from the original two-parameter model. Otherwise the compiler would become a fit engine rather than a decompiler.

The same protocol generalizes to mechanical linkages: a derived continuous path should be checked under interval-certified clearance and exact constraints, not just at finite waypoints. To move beyond molecular mechanics into quantum matter, a domain mapping would also have to derive complex amplitudes, Hilbert-space structure, observable algebra, statistics and their measured constants; none follows from the integer incidence bridge alone.

15. Limitations and corrections retained in the final theory

This synthesis retains rather than hides its principal corrections. The original scalar overlap budget was replaced by an irreducible-by-irreducible capacity bound; rank equality at an invariant reader was shown not to imply closure below family granularity. A reader described as faithful may still discard internal frequency

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information. The price-blind sphere was traced to an existing octahedral core, and its zero-weight faces to a coarse reader, not an absolute cochain-degree obstruction. A wrong finite chain-length bound was replaced by an exact positive-definite compatibility form. The ideal Gray traversal was separated from the self-avoiding motion graph. A 13-of-38 diamond-lattice connectivity report became 16-of-38 when a branch crossing was handled. A proposed uniform Laplacian was shown not uniquely selected by symmetry; the signed gradient and nonnegative conductance were separated as distinct objects. Finally, the numerical RIS agreement across two implementations was separated from experimental agreement.

There are also important archiving limits. Much of the larger historical ledger refers to session-local scripts or hashes whose complete source text is not retained; some claims have been rebuilt and others are provenance-limited [1]. The present paper records results **as reported and audited in the supplied corpus**, except where the displayed formulas provide an independent derivation. Assertions that require a particular source code file, a full prediction seal, or an unseen raw measurement remain conditional on that missing material. The full Nexus metaphysical claim—matter as constraint itself, with no separate external compiler—is a research interpretation. Finite operator correspondences do not settle it.

16. Conclusion

The Nexus programme has reached a materially different stage from a collection of analogies about binary numbers. Its exact objects are now **paired addresses, reversible routes, boundary operators, representation-valued deficiencies, reader channels, topology of permitted continuation, and physical observables**. The key operational discovery is not a particular numeric constant. It is a reproducible *direction of derivation*: start with a relation, read the methods that relation supports, distinguish addressable from executable transformations, and use the missing output's shape to identify the next independent readout.

Three findings make this direction concrete. First, the native palindrome-aware partner ring has an exact, invertible ideal-chain shape reader, including a head-to-tail reversal law and rigid suffix half-turn dictionary. Second, the two-token relation ledger demonstrates that an apparent zero or complete readout may arise from the reader rather than from missing physical or mathematical structure; a deficiency must be typed and closure certified. Third, tetrahedral geometry and an energy-weighted RIS transfer take the same method-first logic into a real physical domain: known polymer rearrangements emerge under declared constraints, and the chain-size calculation becomes quantitative. Its discrepancy with external data—and the rank-two requirement for independent physical parameter identification—is a **scientifically useful unresolved result**, not something to conceal.

The emerging system is best characterized as a **relational decompiler with explicit observability and admissibility contracts**. It offers concrete tools for kinematics, polymer modeling, measurement design and computational verification. Whether those tools can be extended into a uniquely predictive theory of matter will depend on closing the constitutive and quantum links without fitting or inventing them. Until then, the strongest defensible formulation is simple: **what transfers between domains is the shape of the operation—the in-between—not the label or value of the thing being read.**

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Appendix A. Operational contracts for a reusable decompiler

A decompiler intended to transfer this work beyond the original arena should accept (i) a typed configuration domain X; (ii) an explicit local action and inverse or recovery route; (iii) boundary and admissibility predicates; (iv) a relation vocabulary with an exact or controlled numerical basis; (v) a reader with a declared kernel or visibility profile; and (vi) a falsifiable measured target. It should return *separately*: address equivalence, algebraic closure, local infinitesimal mobility, second- or higher-order compatibility, global continuous reachability, and empirical agreement. No one of these certificates is a substitute for another.

For a finite group-stable rational relation space, the execution route is: construct W, calculate $D = A/W$, identify its irreducible content, use the reader character to determine visible deficits, and certify a proposed closing method by an actual span equality. For an ideal physical chain, replace relation-cell rows with bond-length, angle and contact constraints; preserve actual geometric inverses; then check singular branches and clearance. For a material measurement, construct the observable transfer, reduce by symmetry, and compute the rank of the inverse parameter reader **before** interpreting its numerical mismatch.

Appendix B. A correction-aware status ledger

Claim	Ledger status	Reason
Fixed-point-free partner; face/difference inverse	STANDS	Explicit algebraic inverses
Ring-to-ideal-chain difference and reversal	RUNS / STANDS	Exact identities, plus 8,188 finite tests
Self-avoiding half-turn graph connected for all lengths	OPEN	Exhaustive checks end at $m = 12$
Reader-vocabulary representation equation	STANDS	Classical semisimple representation theory
The 235-dimensional witness and its reader spectra	RUNS	Exact source-ledger rational computations
33-orbit 3^- hidden deficiency	RUNS	Witness-specific exact computation in ledger
Boyer endpoint lock / Schatzki rigid turn	RUNS	Ideal-chain constraint audit; literature calibration
16/38 seven-bond-class connectivity	RUNS	Corrected branch-crossing audit
Interval-exact clearance for every reported polymer path	OPEN	Finite-waypoint certificates have stated limits
Polyethylene 72-to-4 RIS algebra	STANDS / RUNS	Exact symmetry reduction; two numerical implementations
413 K model equals observed C_∞	DEAD END	Model 6.2297 versus reported ~7–8 values
Experimental trans-fraction at comparable 413 K conditions	UNLINKED	No independent matched measurement in source record
First-principles constitutive law and general quantum matter	OPEN	Not derived from binary ring construction

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The original sixteen-section synthesis is preserved above. Appendices C–H incorporate subsequent operator details, physical-motion constraints, and a targeted correction audit from the 24 October technical synthesis.

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EXPANDED TECHNICAL APPENDICES · 24 OCTOBER 2026

This edition retains the original mathematical paper and adds the reversible-ring, stream, steric-geometry and inverse-readout machinery supplied in the subsequent technical synthesis. Claims absent independent audit in this revision are attributed to the project ledger; no retrospective result is recast as a sealed prospective physical prediction.

Appendix C. The native ring as an operational machine: pairing, clock, and direct addresses

C.1. Address geometry before interpretation

The expanded October record supplies a useful distinction omitted by a bare list of binary words. At LEN n, the word inventory X_n={0,1}^n is complete, but a word's ring location is determined by an additional, explicitly declared partner and order construction. Let J denote bit reversal and C bitwise complement. The native involution is P(x)=J(x) off the palindromic set and P(x)=C(x) on that set. Its fixed-point-free pairing is a statement about the word-domain operation, not a claim that physical systems are made of binary strings. Each canonical pair contributes one representative to the left ring face and one partner to the opposing face. Ring-index opposition expresses the native facing law R_n[N−1−i]=P(R_n[i]), where N=2^n; the seam exchange is part of the specified address convention.

The construction cannot be simplified into "every canonical representative starts with zero." A canonical pair can have two MSB-one members when the outer bits agree and an interior reversal wins the comparison. The reported exact count of MSB-one canonical representatives is 2^(n−3)−2^⌊(n−3)/2⌋ for n≥3. I independently enumerated the partner map for n=3,…,12 in this revision; the predicted sequence 0,1,2,6,12,28,56,120,240,496 matches the observed counts at every width. This matters operationally: a direct-address reader that assumes one hemisphere equals one MSB value returns incorrect addresses, despite preserving the coarse two-face picture.

The project record develops palindrome-aware rank/unrank formulas and a center-out E/G lift for accessing large rings without materializing 2^n words. The source reports exhaustive low-width checks and direct high-LEN queries, including n=64 and beyond. Those results are valuable to the decompiler because the location is computed from the relation grammar rather than searched in an enumeration. The asymptotic claim of linear-in-width addressing depends on the implementation and arithmetic model; it is not a physical time-complexity measurement.

C.2. One Inverter: a deterministic twisted return

The single-bit continuation without a fixed point is NOT. Applying this local operation only to the bit that exits a cyclic register gives the affine twisted rotation

$$\tau(x_0x_1\ldots x_{n-1}) = x_1x_2\ldots x_{n-1}\,\neg x_0; \qquad \tau^n = C; \qquad \tau^{2n} = I.$$

This is an exact finite operator identity. After n steps the observable word lies on its complementary face; after 2n it returns. At n=2 the orbit 00→01→11→10→00 is a Gray quarter-turn. Thus a closed return in the

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complement quotient need not be a return to the same word. That distinction becomes useful when interpreting a mirrored physical endpoint: quotient closure is weaker than a continuous lift between distinct endpoints.

At width three, P and C cease to be the same permutation, and the four operations {I,P,C,PC} form a Klein four-group on an appropriate four-element orbit. This is an internal symmetry of the ring. Its natural even/odd character channels connect the native ring to the later parity decomposition of ideal-chain shape readers. The group action is an algebraic identification; no independent spatial SO(3) dynamics or quantum spin representation follows merely from its existence.

The reported odd-lane alternating-complement lift Λ_q (q odd) interlaces a smaller s-bit word with its complement in q blocks and satisfies $\tau_{qs}\Lambda_q=\Lambda_q\tau_s$ and $P_{qs}\Lambda_q=\Lambda_qP_s$. I tested the τ and P intertwining equations directly for all $s=1,\ldots,5$ and $q\in\{3,5\}$; they hold. The operation is a precise example of a smaller running ring embedded in a larger address domain while retaining its methods. This is stronger than observing a recurring numerical pattern, but still belongs to finite reversible word dynamics.

C.3. The square-root bridge and a caution about its interpretation

Let T advance one position on the specified partner ring, and form its complement-conjugate $T_C=CTC$. The algebra establishes $PTP=T^{-1}$. The supplemental audit reports a stronger operator comparison: the two permutation matrices T and T_C have no common directed image, so their Frobenius inner product counts zero shared arrows on every audited width. Their difference is orthogonal in the permutation-matrix inner-product sense, not a universal 90-degree angle of physical trajectories.

Define $B_{op}=T_C\,T^{-1}$. The reported bridge gives $B_{op}=S^2$ for the half-step $S=PC\,T^{-1}$. Since C and P commute, the factorization follows by composition using $PTP=T^{-1}$; it is an algebraic square root of a relative permutation, not a unique square root of a physical Hamiltonian. The cycle census records an origin-to-seam spine at widths 2–5 and detached complementary branches starting at n=6. Additional symbolic branch certificates describe 16-state and 28-state rung families at high widths. These provide useful routing and invariant substructures, provided the declared address ordering remains fixed.

Appendix D. Stream duality and the importance of a retained return

D.1. What the difference reader retains

For even n, the partner-fold difference $D(x)=x\oplus P(x)$ is a palindrome and is nonzero. The zero residual is excluded by the native fixed-point-free partner definition; the palindromicity follows because P is either reversal or complement on a palindrome. At n=64 the difference has only 32 independent mirrored positions, even though the original word has 64 positions. The record identifies $2^{32}-1$ distinct nonzero residuals, with a special doubled preimage count at the all-ones residual. This is a concrete quotient: its compressed readout deliberately forgets the face and part of the input distinction.

The inverse is not recovered from the difference alone. As in the pairwise difference law $(a,b)\leftrightarrow(a,a\oplus b)$, one must retain the appropriate reference or phase coordinate. "Way home" in this paper means a specified

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recovery interface or an explicit declaration of information loss, never an assertion that every many-to-one output has a unique inverse.

D.2. Tagged shifts have a complete local inverse

Let x be a 64-bit aperture and write it as x=a‖y, with a the departing two-bit prefix and y the remaining 62 bits. An incoming pair b produces x′=y‖b. The map (x,b)↦x′ is many-to-one if a is discarded, but (x,b)↦(x′,a) is invertible. Its inverse reconstructs x=a‖(x′≫2) and b=x′ mod 4. This is a genuine reversible stream read, independent of the semantic content of the bytes.

In the supplied audit, two named ASCII streams yield 216 two-bit-stride transitions; 169 of those transitions have a different ring jump from the counterfactual transition obtained by appending 00. The outermost unequal mirrored pair determines which orientation wins the canonical comparison, while the innermost unequal mirrored pair determines where an inner palindrome emerges. These are address-selection and quotient-boundary readers, respectively. The observed long jumps show that sequential shifts in an input stream need not be nearest-neighbor steps of T. This prevents the common substitution of input chronology for native ring adjacency.

Appendix E. From the planar grammar to physically executable paths

E.1. A valid address is not a valid kinetic edge

The six-bead lattice-catalog experiment embeds six legal conformations in an eight-address register. A cyclic address-hopping term necessarily crosses the legal/illegal partition, even when the invalid addresses carry high finite penalties. In contrast, the honeycomb representation maps each m-bit turn word directly to the (m+2)-bead ideal chain. The legal subset is selected by self-avoidance on the resulting coordinates. Its definition is a precondition on execution, not an assignment of "invalidity" to a numerical address.

The ideal-chain reader decomposes a turn word into a complementary frame bit and adjacent XOR differences. The differences determine ordered intrinsic distances in the tested model; the frame determines the facing embedding. A native ring partner reverses the differences, corresponding to head-to-tail reversal of the ordered chain shape. Suffix complement operators implement rigid half-turns about selected bonds in three dimensions. These relations give a complete kinematic dictionary for the ideal geometry without asserting that a chosen Gray ring successor is itself an admissible physical trajectory.

E.2. The methods form a graded precondition hierarchy

The supplemental honeycomb audit details a hierarchy beyond the earlier paper's short treatment: end-limited moves, conditional four-turn hexagon rewrites, multihexagon 10-cycle exchanges, and larger end moves act on different restricted move graphs. Algebraic GF(2) mask reachability is necessary but not sufficient: a move must also satisfy its local substring precondition and post-move steric conditions. The zigzag family Z_m forms a closed subdomain under the both2 vocabulary; boundary-accessible 10-cycle methods can remove that particular closure, while a different class of alternating loops survives some

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symmetric extensions. The both3 vocabulary is reported connected at m=4,…,21, with the first small isolated symmetry orbit appearing at m=22; no claim of all-width connectedness follows.

These data are important because they locate an obstruction's owner. If a configuration fails to connect, the cause may be an absent generator, a generator's unmet local precondition, an excluded collision state, a restricted boundary, or a genuinely disconnected continuous configuration space. Reporting only the number of graph components loses that distinction. The decompiler output must include the generator vocabulary and its domain of validity.

E.3. A bypass can be executable without a planar waypoint

The two-control-torsion identity e_j=D_j⊕D_{j+1} expresses one local bit flip as a composition of two physical half-turn operations, but their schedule matters. Executing them serially may force a planar intermediate with a steric collision. A simultaneous path in the control torus $(\phi_1,\phi_2)=(\theta,\pm\theta)$ can avoid that planar corner. The source reports 68 endpoint-valid terminal gates in the m=23–26 residual-lock set, including 12 for which both serial planar intermediates are blocked, and clearance certificates for synchronized three-dimensional schedules.

This is a constructive local mechanism in the declared ideal chain model. A finite verified set of swept moves is not evidence that every physical polymer rearrangement works by the same route. Steric radii, side groups, barriers, dynamic rates, solvents, and the fixed-exterior assumptions must remain attached to any transfer into real chemistry.

Appendix F. Higher-order rigidity and diamond-lattice mirror geometry

F.1. First-order flatness does not decide mobility

At a planar configuration of a spatial linkage, normal displacement columns of the first constraint Jacobian may vanish automatically. A large tangent nullspace can therefore report apparent freedom even when no nontrivial exact path exists. The appropriate next read is the second-order compatibility form: for a first-order tangent z, the quadratic residual q(z) must be orthogonal to every left-null vector of the Jacobian. The 12-cycle collinear crankshaft in the ledger has an explicit positive-definite sum-of-squares combination after factoring out its rigid rotation, closing the nonrigid local branch at second order. This supersedes the earlier invalid span-bound explanation.

A separate contact example has squared-distance residual $c(\theta)=3(1-\cos\theta)$, so its derivative vanishes at $\theta=0$ although its second derivative is 3. For a specifically declared energy $U=(\eta/2)c^2$, the first nonzero restoring term is $(9\eta/8)\theta^4$. That quartic law is conditional on the chosen energy penalty, not an automatic prediction of a real contact potential. Distinguishing contact (one generic scalar constraint) from point welding (three) or full pose docking (six) avoids an artificial rigidity count.

F.2. Boyer and Schatzki are discriminated by allowed motion, not their names

The tetrahedral diamond-lattice model provides a more physically relevant calibration because three fixed torsion states and exact tetrahedral geometry are inputs. An endpoint exchange of the shortest Boyer type

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exists as a pair of conformations but is isolated under strict bond-length, valence-angle and fixed-exterior constraints. The Schatzki-type eight-bead crankshaft has aligned flanking bonds and a proper rigid rotation through 120 degrees. These are recoveries of known polymer mechanisms, not claims to have discovered them or to have deduced all dynamical activation barriers.

The corrected seven-bond audit records 16 connected exchange classes out of 38, not 13. Three missed classes connect through the all-gauche crossing; the correction depends on a verified branch continuation, not an altered count of conformational endpoints. Singular points are associated with consecutive trans torsions and a locally planar zigzag in the enumerated cases. Such local flatness is a necessary condition in the measured finite family, not a sufficient condition or universal theorem about tetrahedral chains.

F.3. Two-sheeted quotient: the exact theorem and the remaining missing link

For the mirror pair with reflection R(x,y,z)=(z,y,x), releasing the specified boundary bond increases local infinitesimal mobility from one to two degrees of freedom. The exact backward-heading calculation finds no configuration fixed by R in the extended constrained space; its smallest squared endpoint mismatch in the mirror-fixed plane is 8/3. Therefore R acts freely and the quotient by the two-element reflection group is an unbranched two-sheeted cover, under the declared smooth configuration-space assumptions.

The conclusion is intentionally limited. A free involution does not imply that A and RA belong to the same connected component. Their connection exists precisely if a loop in the quotient has an odd-monodromy lift starting at A and ending at RA. The twisted binary clock demonstrates the abstract *type* of half-return, but does not provide the missing physical loop. The 50-step constrained interpolation failing to reach its target does not prove the opposite, either. The physical odd-loop return remains OPEN. This preserves the shape of the unresolved relation instead of substituting another trajectory search for it.

Appendix G. Quantitative material response and the falsifiable boundary

G.1. A reduction is not a force law

The ring-to-spring bridge uses the integer incidence row B=[−1,1]. Its evaluation over F_2 is XOR; over the real numbers it measures extension x_1-x_0. A translation-invariant symmetric two-site quadratic stiffness is proportional to B^TB; once an energy scale and masses are supplied, its normal-mode structure follows. No nonzero additive embedding $F_2\to R$ identifies the respective field elements. What transfers is the incidence relation and its compositional algebra, not the physical units, energy scale, or material chemistry.

The polyethylene RIS calculation extends the same method to a larger readout. The declared tetrahedral continuation grammar and the gauche-sign reflection symmetry reduce the spatial correlation problem from 72 states to a four-scalar resolvent. The source gives the exact independent-torsion result $C_\infty=2(2+\sigma)/(3\sigma)$, including the unweighted case $C_\infty=2$, and a full-model crosscheck at numerical precision. These validate the reduction for its input model. They do not independently establish that real polyethylene uses ideal tetrahedral bond angles or precisely three discrete torsional minima.

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G.2. The measured gap cannot be renamed as a landing

With externally sourced illustrative energies E_g=2.1 kJ/mol and E_p=8.4 kJ/mol at 413 K, the declared reduced model gives C∞=6.229676328. The opposite-gauche-forbidden *limit* at the same E_g gives C∞=6.715345190. The earlier published scattering central benchmark is about 7.8, with nonidentical measurement methods and thermodynamic assumptions. The calculated 6.715345190 is a proved value at ω=0, but calling it the global maximum for 0≤ω≤1 is stronger than the available evidence: the supplemental record reports a strictly ordered 1,001-point sweep, which supports monotonicity numerically but does not itself prove it analytically.

Even if full monotonicity were established, the empirical discrepancy would remain a property of the *specified ideal-lattice energy model*, not a demonstration that the ring has failed or that it fixes real chemical energies. Flexible bond angles, torsion minima, and non-nearest-neighbor interactions are possible missing physical dependencies; none should be silently inserted until independently constrained.

G.3. The inverse gap is a second independent measurement

The parameter pair (E_g,E_p) cannot be uniquely reconstructed from one chain-size observable. The two-output reader (C∞,p_trans) has a calculated local Jacobian, in parameters measured in kJ/mol,

$$J \approx [[0.790254, 0.134369], [0.052463, 0.003102]], \quad \det(J) \approx -0.004598.$$

Its nonzero determinant supplies local mathematical identifiability in the two-parameter RIS model, while the approximately 140 condition number warns that even modest experimental errors can be amplified. The independently matched trans fraction has not been supplied for the 413 K material condition; it is UNLINKED, not replaced by a value from a different phase of polyethylene.

The first prospective physical test must therefore seal *both* predicted readouts, preserve the energy-parameter source, temperature, uncertainty model, and phase specification, and identify the way home through the independent 72-state implementation. A second experimental number should decide the two-parameter prediction before any extension of the constitutive law is allowed.

Appendix H. Practical decompiler workflow and status discipline

The complete operational workflow is: (1) declare the native domain and generator vocabulary; (2) compute the route product and inverse or record a quotient kernel; (3) separate available algebraic masks from executable moves with preconditions and boundaries; (4) reduce by exact symmetries before enumerating trajectories; (5) find the deficiency or rank-nullity obstruction; (6) request an independent reader only when the deficiency is AMBIGUOUS; and (7) seal a falsifiable quantitative readout before importing target measurements. A route that reaches the correct *type* of structure but not its material realization remains OPEN, even when its mathematics STANDS.

The paper's practical implications are consequently specific. In molecular kinematics, it can return methods that are excluded by a boundary clamp, permitted by an exact rigid sweep, or unresolved due to a nonrigid motion component. In mechanism design, it can distinguish first-order zero modes from executable flexes

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and identify the missing higher-order constraint. In sensor design, it can recommend which symmetry sector a measurement must resolve. In computational verification, it can distinguish a reversible internal state route from a many-to-one published output. In polymer spectroscopy, it can select observables that remove parameter ambiguity before any fitting campaign. In every case, the decompiler is useful only to the extent that the chosen operator relations genuinely describe the physical domain.

Added-check ledger for this expanded edition. STANDS: fixed-point-free partner involution, twisted clock identities, boundary operator across coefficient systems, quotient-cover/odd-monodromy criterion, and local inverse-function rank test. RUNS (independently checked during this revision): the MSB-one canonical representative counts for $n=3\ldots12$, and the odd-lane P/τ intertwining identities for $s=1\ldots5$ and $q=3,5$. RUNS (as recorded in the supplied project audit, not all independently reconstructed in this revision): the larger honeycomb connectivity census, reported clearance certificates, the 16/38 diamond exchange correction, and the 72-to-4 transfer-matrix consistency tests. DEAD END: treating two-sheeted quotient topology as proof of a realized physical connection; treating a finite monotonicity sweep as a rigorous global extremum proof; and treating a correct mathematical reduction as a first-principles constitutive law. OPEN: an exact physical odd-loop lift for the reflected diamond configuration and an independently linked, prospective two-observable material prediction.

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